Code No: 136FC **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, February/March - 2022
FUNDAMENTALS OF ROBOTICS
(Common to CE, ECE, CSE, IT, AE)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Sketch and explain the four basic robot configurations classified according to the coordinate system.
b) Differentiate CAD/CAM and robotics. [9+6]

2.a) Explain the relationship of robotics with industrial automation and illustrate the same with a suitable example.
b) Give the applications of robot with respect to Cartesian, cylindrical and polar arm configuration. [7+8]

3.a) Discuss about Vaccum Grippers along with their advantages and disadvantages.
b) What are the various factors in selection and design of end effectors? Explain. [8+7]

4.a) Discuss in detail about Magnetic gripper with neat sketch.
b) What do you understand by degree of freedom (DOF)? How many DOFs are required to position an end effector at any point in 3-D space? [8+7]

5.a) Determine the transformation matrix T that represents a translation of 'a' units along x-axis, followed by a rotation of β about x-axis and followed by a rotation of θ about z-axis.
b) What is homogenous transformation matrix? Explain four sub matrices. [8+7]

6.a) What is the role of D-H notation? Explain their importance in solving Forward Kinematics.
b) Write homogenous transformation matrices for rotation in 3D. [9+6]

7.a) What are the common types of motion that a robot manipulator can make in travelling from point to point?
b) A point to point robot with a revolute joint moving with velocity of 15 deg/sec, traverses from an initial position of 12^0 to a final position of 60 deg/sec. Determine the position and velocity at the end of 1, 2 and 3 seconds. The range of initial and final position is covered in 6 seconds with a finite acceleration of 8 deg/sec^2. [8+7]

8.a) Discuss the relative merits and demerits of different textual robot languages.
b) Explain different program instructions of teach pendant method with suitable examples. [8+7]

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